

Parking Management System

Requirements Definition Document

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1 Introduction

In an age where people commute daily between workplaces, schools, and commercial centers, effective parking regulation has become essential to modern transportation systems. Clearly defined parking standards help ensure that vehicles are managed efficiently, reducing congestion, minimizing wasted time, and improving the overall reliability of driving. Having well-regulated parking supports economic activity by allowing businesses and institutions to accommodate visitors smoothly, which encourages consistent traffic flow and accessibility. By providing structured, predictable parking solutions, these systems contribute to a more dependable commuting experience while supporting the broader economy that relies on safe and efficient mobility.

A multistory Parking Management System (PMS) is used to help people effectively find and use parking in spaces where reducing congestion, confusion, traffic, and wasted time is needed. Through the monitoring of available spaces and controlling vehicle entry and exit, the system improves safety and traffic flow while maximizing the use of limited space, particularly in urban spaces. This system provides a more organized, convenient, and reliable parking experience for drivers in busy areas like college campuses, hospitals, and big shopping centers/plazas.

This project presents the design of a multistory Parking Management System. The system simulates the operation of a software-controlled parking facility by managing vehicle entry, parking availability, and exit, taking into account multiple floors. It maintains accurate tracking of available parking spaces at both the floor and overall structure levels while enforcing capacity constraints and notifying potential parkers of the available spots. Through consistent state management, the system models how vehicles are admitted, parked, and removed from the structure, providing a realistic representation of parking structure operations. This provides potential parkers with a luxurious user experience that is economically focused.

2 Definition of Terms

In this section, we define some relevant terms that are used frequently throughout the paper for accurate communication. The reader can use this section as a future reference while going through the document.

Administrator - Generally, owners of the structure or specifically personnel in charge of managing the PMS, enabling manual override of certain functionality and manually updating parking space availability whenever necessary.

Availability - The status of a parking spot, which will be either filled (vehicle parked) or available (no vehicle parked).

Availability Display - The required main display is made available in front of the parking lot to provide availability counts for various spot types. Additionally can refer to displays on each floor of the lot displaying the same information as the aforementioned main display.

Black Box Diagram - A diagram used to represent at a high level the required input and expected output from a system.

Client - Individuals who drive a vehicle and desire to use a parking space for their vehicle.

EV - Electric Vehicle. Requires a designated charging station to recharge.

Handicapped - Parking space for individuals who have a condition that restricts ease of movement. Can also refer to as an *accessibility space*.

In-Transit - A status of a vehicle referring to when the vehicle is within the garage, but not parked. Additionally, the count of these is the number of vehicles with this status.

LEDs - A light-emitting diode, used for indicating parking space availability.

PMS - The Parking Management System, consisting of all the devices and software necessary to manage and oversee parking lot operations.

Spikes (Spiked Speed Bump) - A traffic-calming device used to direct traffic to one direction using spikes and a raised bump. Occasionally referred to as *directional spikes*.

3 Objectives

PMS aims to provide three essential objectives aimed at providing administrators with a functioning system. The general goal of PMS can be summarized as an adaptable, sensor-based system configured to a parking structure providing clear and accurate data regarding parking availability. The three objectives are declared as Accuracy and Data Transparency, Accessibility, and Flexibility/Ease of Installation. The goals of PMS are detailed below.

3.1 Accuracy and Data Transparency.

PMS provides sensors and displays that must provide accurate, real-time data to our clients regarding the current availability of parking spots in the parking structure. Moreover, the system must also regularly update displays of data to clients regarding the currently detected vehicles within the parking structure to ensure that clients are informed and updated about other clients attempting to secure parking spots as well. Administrators should be able to pinpoint the current activity regarding individual parking spots; manual, administrator-based changes to sensor configurations must be reflected on displays, regarding parking availability upon confirmation of configurations when needed. Regarding availability updates, the entrance gates must be signaled promptly to alert clients of an eventual closure due to unavailable parking spots. Visual data regarding parking availability and vehicles must be present to clients upon every level of the structure, both the structure as a whole and for each level.

3.2 Accessibility

PMS must provide extensive configurations aside from default system configurations - to allow administrators to adapt the system to their parking structure's needs. Inclusivity and environmental concern are imperative with reference to Handicapped and EV spots; hence, configurable visual settings, such as for sensor lights and displays, must be offered. These settings must allow for easily-identifiable, visual cues in sensor light or displays to alert clients of the specialized parking spots available in the structure. Additionally, the PMS offers specialized sensors for parking spots to accommodate Motorcycle vehicles that have a differing geometry than conventional vehicles, like cars and trucks. Devices regarding PMS and its park structure information should be visually distinguishable in order to draw the client's attention. Ultimately, PMS enhances an economical management system which is free of charge to clients; PMS does not charge for parking itself.

3.3 Flexibility/Ease of Installation

PMS must not hinder administrators in installation and maintenance. Sensors and gates for parking spots will be running on an interconnected network system that should allow for communication between the devices and the PMS. The sensors themselves must be configured right next to each parking spot in such a way that parking is not obstructed; the bulk of sensor technology must not be distracting to drivers.

4 System Organization

The system organization requirements of the PMS are outlined as follows, beginning with a general overview of the floor of a typical parking lot, with all system requirements. Further, this section will outline in more detail the interface expectations of the systems, detailing what is expected to be handled and the necessary outputs based upon these inputs.

4.1 Overview of Parking Lot

The overview of a typical lot, as seen in *Figure 4.1*, showcases the necessary components for the PMS to interface with, offering a visual aid as to what will be necessary for the PMS to function, along with placement requirements. Note that the figure represents a single ground-floor layout, though the PMS will have no presumption of the number of floors and will account for multiple.

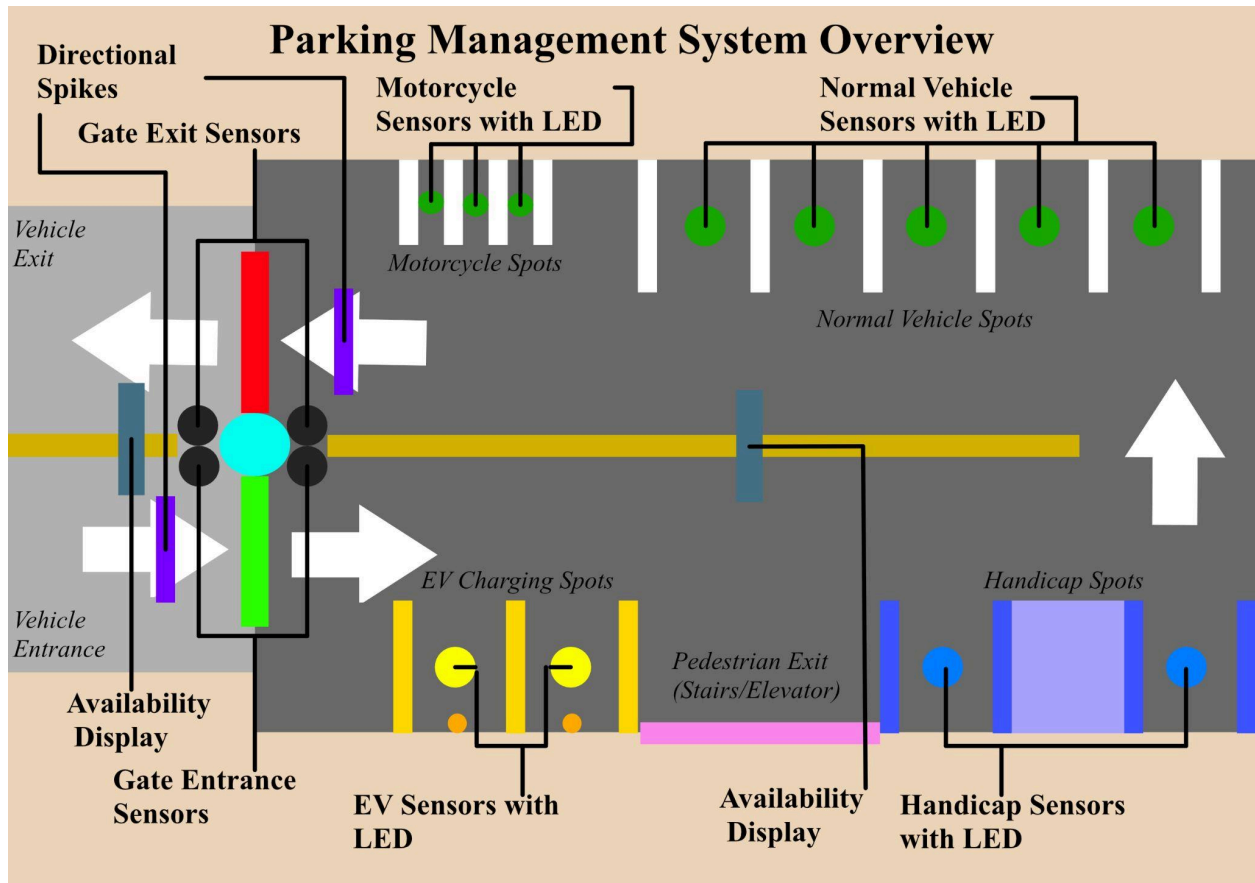


Figure 4.1

A general overview of a typical parking lot that includes the hardware needed to track all requirements of the PMS.

4.2 PMS Black Box Diagram

To give further detail and visualization on how the PMS will utilize the components seen in *Figure 4.1*, the following Black Box Diagram in *Figure 4.2* is used to represent the input as is to be expected from the system hardware components in *Figure 4.1* and the resulting output that will be required of the system.

Black Box Diagram

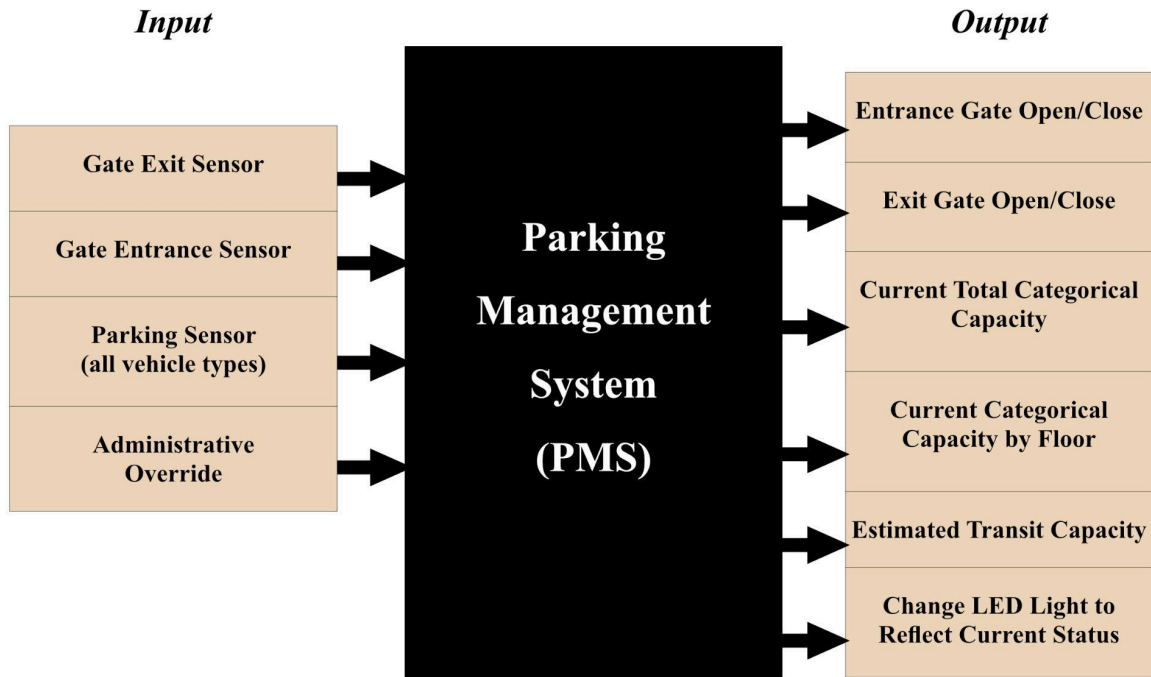


Figure 4.2

A black box diagram of what the PMS must be able to interface with (input) and what outputs the system must be able to track according to requirements.

The following section discusses with more specificity the interfaces within this diagram to clearly define expectations.

5 Interfaces

The environment that the PMS must operate in can be described by a set of interfaces. This set can be further divided into subsets of input and output interfaces.

5.1 Input Interfaces

The input interfaces, as seen in *Figure 4.2*, will provide signals to the PMS. Following is a description of each of those components.

5.1.1 Gate Exit Sensor

The gate exit sensor will be located at the exit of the parking lot. This sensor will automatically detect the approach of a vehicle, ignoring false stimuli (pedestrians, bicycles, etc.). The exit gate will then open and will stay open until the vehicle is completely clear of the gate as a safety

mechanism. This sensor will interface with the PMS to notify of an exit request and verify success. Additionally, a spiked speed bump will be in place to prevent backtracking.

5.1.2 Gate Entrance Sensor

The gate entrance sensor will be located at the entrance of the parking lot. This sensor will automatically sense when a vehicle is approaching the gate, while likewise ignoring false stimuli. As a vehicle approaches, the gate will open only if there is availability in the lot, interfacing with the PMS to notify of the potential entrance of a vehicle. Again, as a safety mechanism, the gate entrance sensor will also track the vehicle in order to ensure that the gate does not close until the vehicle is completely clear of the gate. The gate sensor will again interface with the PMS on a successful entry. A spiked speed bump will also be in place to prevent backtracking.

5.1.3 Parking Sensor

A parking sensor will be included and identified with every spot. The sensor will signal the PMS that a vehicle has filled and left a spot. Further, associated with each sensor is an LED light indicating the availability of the spot.

5.1.4 Administrative Override

The administrative override will be a hosted and provided application to interface with the sensors/LEDs of the lots directly, changing the state manually of a space as needed. This capability is provided as a safety and user-friendly measure, considering the potential for needing to shut down sections of the lot in cases of damage, maintenance, or necessary resets.

The PMS will receive instructions through this administrative application and will be able to balance those requests with the automatic signalling of sensors and carry out tasks accordingly.

5.2 Output Interfaces

The output interfaces, as seen in *Figure 3.2*, will receive signals from the PMS. The PMS will also be tracking and handling any information required of these interfaces to adhere to the strict practices of reliability and accuracy within the system. Following is a description of each of those components.

5.2.1 Entrance Gate Open/Close

The entrance gate will be controlled through instructions issued by the PMS according to gate entrance sensor signals. Further, the PMS will disable the entrance gate based on availability within the lot, allowing for clear availability communication for clients.

5.2.2 Exit Gate Open/Close

The exit gate will likewise be controlled through instructions issued by the PMS according to the exit sensor signals. This gate will never be locked due to caution in emergency scenarios.

5.2.3 Current Total Categorical Capacity

Based upon a combination of entrance/exit signals and parking spot sensor signals, the PMS will carefully track and output to availability displays the total free parking spots in the entire lot, including special categories of availability. The special categories the PMS will make available are Handicap, EV charging, and Motorcycle spots.

5.2.4 Current Categorical Capacity by Floor

Based upon a combination of entrance/exit signals and parking spot sensor signals, the PMS will be tracking total and special categories (as mentioned in 5.2.3) by each floor to output to availability displays.

5.2.5 Current Estimated Transit Capacity

The PMS will also include tracking of cars in transit. This information will be made available on the availability displays as a measure of convenience and accuracy in availability metrics for clients.

5.2.6 Change LED Light to Reflect Current Status

The PMS will emit instructions based on various input signals from sensors or administrative control to change the light as feedback to indicate availability. Further, other indications of unavailability based on administrative need will be issued through the PMS as needed.

6 Capabilities

Capabilities for PMS correspond to the functional requirements imposed on the system to ensure parking information services are maintained within PMS. The PMS capabilities ensure that the system is, for the majority, autonomous, with limited requirements for manual intervention. Three categories encompass the main capabilities of PMS: Real-Time Updates Capabilities, Pedestrian and Vehicle Differentiation Capabilities, and Adaptable System Configuration Capabilities. Details of the PMS capabilities are presented below.

6.1 Real-Time Updates Capabilities

1. The sensors should constantly monitor their parking spot to ensure a parking spot is empty/full - updating their LED lights appropriately to the state.
2. The availability displays should be able to update automatically upon every new sensory event detected by parking spot sensors.

3. The in-transit-count displays should be able to update automatically upon every new sensory event detected by entrance/exit sensors.
4. Manual triggering of sensor states (on/off and LED color) may only be induced through non-client interfaces - provided to the PMS administrators.
5. The entrance gate should be signaled upon the system's detection of total unavailability of parking spaces provided from parking spot sensor information, to be able to alert incoming vehicles of an eventual closure due to limited availability.
6. Entrance and exit gates must detect the potential entering/exiting of a vehicle to warn of directional spikes to enforce vehicles to maintain a singular direction, hence, enforcing the in-transit-count detection portion of the system

6.2 Pedestrian and Vehicle Differentiation Capabilities

1. Sensors in each parking spot will tell apart differing vehicles based on the type of parking spot (Motorcycle/Non-Motorcycle vehicles).
2. Sensors in each parking spot will tell apart vehicles and humans, to signal a parking spot as occupied/vacant.
3. Entrance and exit gates may only detect vehicles entering and exiting.

6.3 Adaptable System Configuration Capabilities

1. Sensors in parking spots may be configured by administrators to distinguish Motorcycle/Non-Motorcycle vehicles to compensate for geometric differences.
2. Sensors in parking spots may have their LED lights configured by administrators to different colors to signify particular types of parking spots - such as Motorcycle, Handicap, or EV sections.
3. Parking spot availability displays may be configured by administrators to display counts next to particular types of parking spots signified by representative symbols for clarity on availability information.

7 Design Constraints

The design constraints, as follows, are to be adhered to carefully in the development and consideration of the PMS.

7.1 Administrator Constraints

There are specific administrative constraints that must be accounted for in the PMS. The essential actionable requirements that must be met are as follows:

- Total capacity must be tracked.
- Capacity by floor must be tracked.
- Real-time parking information and availability must be displayed.
- PMS must maintain consistency and accuracy under capacity constraints and actions, specifically entering, parking, and exiting.

- PMS must be able to work with a colored light indication to show available or filled status for each spot in the lot.

The following additional distinctions and requests have also been specified by the administration:

- There will be more than 1 floor within a parking lot.
- Each floor has a fixed number of spots.
- There is 1 exit and 1 entrance.
- There is no payment or permitting required for the parking lot.
- There is no assigned parking required for the parking lot.
- The parking lot must have Motorcycle spots and EV charging spots accounted for.
- As the American Disability Association regulations specify, the parking lot will have at least 1 Handicap accessible parking spot [1]. These spots must also be accounted for.
- Availability tracking must account for in-transit vehicles.
- A user-friendly interface must be provided to allow for administrative override over actions typically automatically executed by the PMS.

7.2 Environmental Constraints

The PMS must be able to:

- Account for any environmental extremes to ensure that all components are functioning properly. Specifically:
 - The sensors must not be impacted by inclement weather (rain, fog, snow, etc.)
 - The sensors must work in both extreme cold and heat.
- Maintain a consistent state through a power outage to provide the reliability expected by the administration.
- Not be impeded by materials in which the lot is made out of (primarily concrete).

8 Conclusion

With the emphasis and importance of reliable infrastructure to support commuters in the modern era, requirements laid out for the PMS have been specially tailored to enable a smooth experience for both drivers and administrators alike. With a focus on accuracy and reliability, the PMS will take the worry out of every morning journey for people of all walks of life, keeping all diverse methods of transportation in mind. Further, special requirements laid out keep the administration in mind, keeping the PMS solid, reliable, and adaptable in the long-term. This system will have the utmost focus on keeping life simpler for all in their daily routines. Through this document, we have clearly outlined the requirements set by the administrator, along with non-functional constraints, and specified the system to meet all said requirements.

9 References

- [1] “Accessible parking spaces,” ADA.gov, <https://www.ada.gov/topics/parking/>. Accessed Jan. 27, 2026.